Calendar-Arbitrage Prevention

Convex order, integrated quantiles, and a model-agnostic soft hinge

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Abstract

Butterfly-freedom keeps a single smile a valid density; calendar-freedom keeps a *surface* consistent across maturities. This note documents how the Vol-Fitter enforces it. With every expiry normalized by its forward, absence of calendar arbitrage is the statement that the terminal laws are increasing in convex order — equivalently, that normalized call prices, and hence total implied variance at each fixed log-moneyness, are non-decreasing in maturity. We give two enforcements. For LQD the convex order is exactly an ordering of the *integrated upper-quantile* curves, which each built slice already carries as its asset-share integral, so the constraint is an elementwise array comparison with no quantile inversion; expiries are fitted nearest-to-farthest under a soft slack. For the SVI and Multi-Core SIV overlays, which carry no quantile curve, a *model-agnostic* hinge enforces $w_{\mathrm{far}}(k) \geq w_{\mathrm{near}}(k)$ directly on total variance. We then explain the subtle but important *confinement* fix — the floor must be sampled only on the traded strike range, or a steep short-dated slice's linear-wing extrapolation manufactures a phantom violation that flattens the in-data fit. A worked example takes an inverted-wing pair from a calendar violation of 0.0059 to 4.58e-07.

Contents

1 What calendar-freedom means

Fix log-moneyness k and let $C_T(k)$ be the forward-normalized call at expiry T. With all expiries normalized to unit forward, the absence of a calendar spread arbitrage is

$$C_{T_i}(k) \le C_{T_j}(k) \quad \text{for } T_i < T_j,\ \forall k, \tag{1}$$

i.e. a longer-dated call is never cheaper than a shorter-dated one at the same moneyness. Because the Black price is increasing in total variance, under deterministic proportional carry (1) is equivalent to total variance being non-decreasing in maturity, $w_{T_i}(k) \le w_{T_j}(k)$. The sharper, model-independent statement is in terms of the terminal laws: writing $Y_T = S_T/F_T$ (mean one), (1) holds iff

$$Y_{T_i} \le_{cx} Y_{T_j}, \tag{2}$$

the Y's are increasing in *convex order*. This is the object the LQD enforcement uses directly.

Heuristic

Convex order with equal means says the longer-maturity law is a "mean-preserving spread" of the shorter one: same forward, more dispersion. That is exactly what a diffusion does — variance accumulates with time — so calendar-freedom is the statement that the surface is consistent with *some* martingale, even if the fitter never names one. Enforcing it on the fitted slices is what stops two independently-fitted expiries from implying a negative forward variance between them.

2 The LQD-exact enforcement

2.1 Convex order as integrated quantiles

For equal-mean laws, $Y_{T_i} \le_{cx} Y_{T_j}$ holds iff the *integrated upper-quantile* curves are ordered:

$$G_i(\alpha) := \int_\alpha^1 Q_{Y,T_i}(u)\,\mathrm{d}u \ \le\ G_j(\alpha) \quad \forall \alpha \in [0,1], \tag{3}$$

with equality at $\alpha = 0$ (equal means). Now recall (Note 01) that for an LQD slice $Q_Y(u) = e^{Q(u)}$ and, in the logit coordinate, $\mathrm{d}u = u(1-u)\,\mathrm{d}z$, so

$$G(\alpha) = \int_\alpha^1 e^{Q(u)}\,\mathrm{d}u = \int_{z_\alpha}^\infty e^{Q(z)}\Lambda(z)(1-\Lambda(z))\,\mathrm{d}z = A(z_\alpha), \tag{4}$$

which is *exactly* the asset-share integral $A(z)$ every built slice already carries on the shared quadrature grid. The calendar constraint is therefore an elementwise comparison of two `a_z` arrays — no quantile inversion, no per-strike call comparison — and, because the grid is uniform in z, it places constraint points densely in the wings exactly where they matter.

2.2 Sequential fitting with slack

The surface is built nearest-to-farthest. The constraint subgrid takes every 25th node (~ 320 points on the 8001-node grid). Each new slice is fitted subject to the soft floor

$$A_j(z_m) \;\geq\; A_i(z_m) - \tau_m, \tag{5}$$

implemented as the squared-hinge slack $\sqrt{W_{\mathrm{cal}}}\,\max(A_i(z_m) - A_j(z_m), 0)$ in the LQD objective (weight `calendarWeight`, default 10^6). The floor is keyed on the constraint *z-coordinates*, so the next slice can enforce it by Hermite-evaluating its own curve there even when it is calibrated on a coarser optimization grid — bit-for-bit equivalent to the node-indexed form on the native grid.

3 The model-agnostic enforcement

The SVI and Multi-Core SIV overlays carry no integrated-quantile curve, so they use Gatheral's equivalent *surface* condition directly: total implied variance non-decreasing in maturity,

$$w_{\mathrm{far}}(k) \;\geq\; w_{\mathrm{near}}(k) \quad \forall k, \tag{6}$$

enforced as a soft hinge $\sqrt{W_{\mathrm{cal}}}\,\max(w_{\mathrm{near}}(k) - w_{\mathrm{far}}(k), 0)$ on a log-moneyness grid. This reads only the previous slice's implied_w(k), so it applies to *any* model — raw SVI, Multi-Core SIV, or the local-vol overlay — and gives the whole surface a uniform calendar treatment regardless of which model draws each slice. As residual rows it is one line, the same squared-hinge shape as the band objective (Note 07):

Listing 1: The model-agnostic calendar hinge (6) (distilled from `calib/calendar.py`).

```
def calendar_residual(w_far, w_near, weight):     # w_* = implied_w(k) on the SAME grid
    return np.sqrt(weight) * np.maximum(w_near - w_far, 0.0)   # 0 unless the later dips below
```

3.1 The confinement fix

Where the floor (6) is *sampled* turns out to be critical.

Caution

If the floor grid is fixed and wide (say $k \in [-1, 1]$, far beyond the traded strikes), then for a family with *linear wings* (SVI) a steep short-dated slice extrapolates to far higher wing variance than a flatter long-dated one, so $w_{\mathrm{near}} > w_{\mathrm{far}}$ out in the untraded wing — a *phantom* violation created purely by extrapolation. The optimizer then flattens the long-dated wing to "cure" a problem the data never posed. This was observed live on NVDA and SPY overlays.

The fix is *confinement*: `variance_floor_grid_from` builds the floor grid on the *observed* quote range $[\min k, \max k]$ only (empty quotes fall back to the wide grid) — a two-line change that is the whole cure:

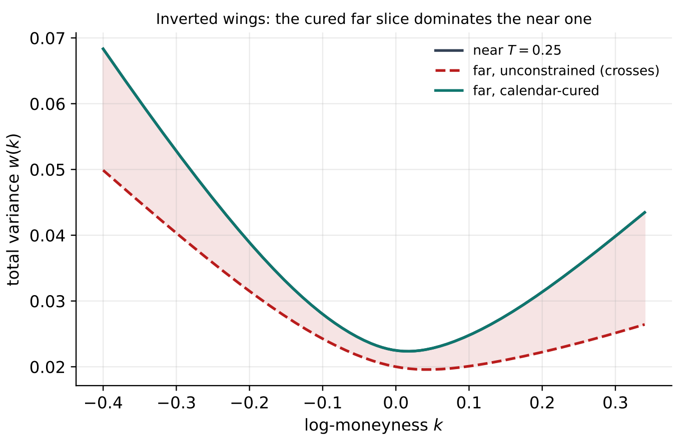
Listing 2: The confinement fix: sample the floor only on the traded range (distilled from `calib/calendar.py`).

```python
def floor_grid(k_quotes, n=41):            # the grid the calendar hinge is sampled on
    return np.linspace(k_quotes.min(), k_quotes.max(), n)   # traded span, NOT a fixed wide grid
```

Calendar arbitrage is only meaningful where prices are observable; confining the floor keeps the constraint where the fit and the data both live, and leaves each model's wings to its own Lee-admissible extrapolation (Note 09). This is the same lesson as the local-vol convex-wing regression of Note 04: *a no-arbitrage constraint must not be sampled where only extrapolation lives.*

4 Worked example

Figure 1 fits an *inverted-wing* pair: a high-vol short expiry ($T = 0.25$) over a calmer long expiry ($T = 0.5$), the kind of term structure an imminent event produces. Fitted independently, the long slice's total variance dips *below* the short one's in the wings (the shaded region) — a calendar violation of 0.0059. Refitting the long slice under the calendar floor from the short one lifts it to dominate everywhere, driving the violation to $4.58e-07$ (numerically zero). Section 4 shows the cured pair's integrated upper-quantile curves correctly ordered, $G_{\text{far}} \geq G_{\text{near}}$ — the convex order (2) made manifest.



(a) total variance: crossing, then cured

(b) integrated quantiles, ordered

Figure 1: An inverted-wing pair: the unconstrained long slice crosses the short one (violation 0.0059); the calendar floor cures it (violation $4.58e-07$).

5 What is original here

The convex-order characterization is classical (Strassen, Carr–Madan); the contributions are implementation. Recognizing that the LQD asset-share integral $A(z)$ *is* the integrated upper-quantile curve $G(\alpha)$ makes the calendar constraint a free elementwise comparison rather than a dense grid of inverted call inequalities. Extending the same no-arbitrage to the SVI/SIV/LV overlays through the model-agnostic total-variance floor (6) gives the whole surface one calendar treatment. And the *confinement* fix (3.1) — sampling the floor only where data lives — is a genuinely non-obvious lesson that the same wide-grid mistake recurs in two places (here and the local-vol convex wing) and is cured the same way.

A Hyperparameter atlas

Table 1: Calendar-arbitrage hyperparameters.

Knob	Default	Role
Surfaced		
`enforceCalendar`	on	Master toggle for calendar coupling.
`calendarWeight` W_{cal}	10^6	Soft-constraint penalty strength.
Hidden		
`CAL_STRIDE`	25	LQD constraint subgrid stride (~ 320 points).
`VAR_FLOOR_KMIN/KMAX`	$[-1, 1]$	Fallback wide floor grid (empty quotes only).
`VAR_FLOOR_N`	161	Wide floor grid resolution.
`VAR_FLOOR_N_DATA`	41	Confined floor grid resolution.
floor grid	traded range	`variance_floor_grid_from`: confined to $[\min k, \max k]$.

Performance. The LQD constraint is an array slice and subtraction on the already-built asset share — effectively free; its Jacobian rows are $-\partial A/\partial\theta$ on the active constraints, supplied by the analytic Jacobian (Note 01). The model-agnostic floor is ~ 40 arithmetic evaluations of the previous slice's $w(k)$. Confinement *reduces* work by not over-sampling the extrapolation region.

References

[1] V. Strassen. The existence of probability measures with given marginals. *Ann. Math. Statist.*, 36(2):423–439, 1965.

[2] P. Carr and D. Madan. A note on sufficient conditions for no arbitrage. *Finance Research Letters*, 2(3):125–130, 2005.

[3] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.